④ $\vec{r}(t) = \langle \overbrace{e^t \cos t}^{x}, \overbrace{e^t \sin t}^{y}, \overbrace{e^t}^{z} \rangle$ eğrisinin $0 \le t \le 1$ aralığındaki uzunluğunu veren integral aşağıdakilerden hangisidir?

a) $\int_0^1 \sqrt{2}\, e^t \, dt$ **b)** $\int_0^1 \sqrt{3}\, e^t \, dt$ c) $\int_0^1 e^t \sin t \cos t \, dt$

$$S = \int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

$x = e^t \cos t \rightarrow \frac{dx}{dt} = e^t \cdot \cos t - e^t \cdot \sin t \rightarrow \left(\frac{dx}{dt}\right)^2 = \overbrace{e^{2t}}\cos^2 t + e^{2t}\sin^2 t - 2e^{2t}\cos t \sin t$

$y = e^t \sin t \rightarrow \frac{dy}{dt} = e^t \sin t + e^t \cos t \rightarrow \left(\frac{dy}{dt}\right)^2 = \overbrace{e^{2t}}\sin^2 t + e^{2t}\cos^2 t + 2e^{2t}\sin t \cos t$

$z = e^t \rightarrow \frac{dz}{dt} = e^t \rightarrow \left(\frac{dz}{dt}\right)^2 = e^{2t}$

$+$

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} = \sqrt{3e^{2t}} = \sqrt{3}\cdot e^t$$

$$S = \int_0^1 \sqrt{3}\, e^t \, dt = \sqrt{3}\, e^t \Big|_0^1 = \sqrt{3}\, e - \sqrt{3} = \sqrt{3}(e-1)$$

e) $[1, +\infty)$

③ $\vec{r}(t) = \overbrace{(3t^2+1)}^{x}\vec{i} + \overbrace{(4t^3-3)}^{y}\vec{j} + \overbrace{(6t+2)}^{z}\vec{k}$ eğrisinin $(4,1,8)$ noktasındaki teğet doğrusunun denklemi aşağıdakilerden hangisidir?

Nokta: $(4,1,8)$ → $x_0\ y_0\ z_0$

Yön (paralel) vektör: $\langle a,b,c \rangle$? → Teğet vektör!

$(4,1,8) \rightarrow t = ?$

$\vec{r}' = \langle 6t, 12t^2, 6 \rangle$

$\downarrow t=1$

$\vec{r}'(1) = \vec{v} = \langle 6, 12, 6 \rangle$

$a\ b\ c$

$4 = 3t^2+1$

$1 = 4t^3-3$

$8 = 6t+2$

$\Big\} \ t=1$

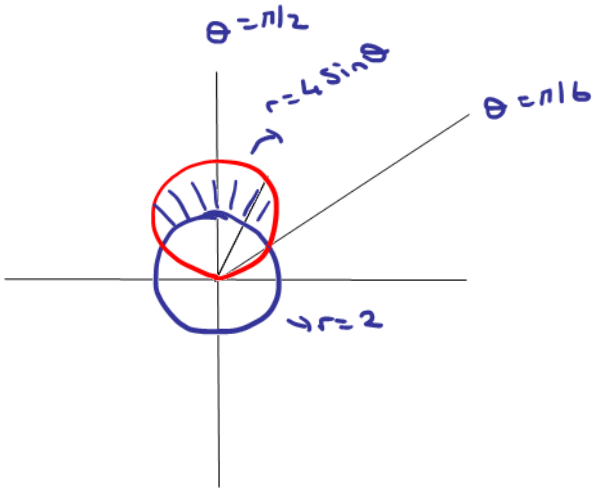
$x = x_0 + at = 4 + 6t$

$y = y_0 + bt = 1 + 12t$

$z = z_0 + ct = 8 + 6t$

$=$

(4) $x^2+y^2=4$ çemberinin dışında ve $x^2+y^2-4y=0$ çemberinin içinde kalan bölgenin alanı?



$x=r\cos\theta$

$y=r\sin\theta$

$dxdy=r\,dr\,d\theta$

$x^2+y^2=r^2$

* $x^2+y^2=4$

$r^2=4 \rightarrow r=2$

* $x^2+y^2=4y$

$r^2=4r\sin\theta$

$r=4\sin\theta$

$r=2 \leftrightarrow r=4\sin\theta$

$4\sin\theta=2 \rightarrow \sin\theta=\frac{1}{2}$

$\theta=\frac{\pi}{6}$

$$A=\iint_D dx\,dy=2\int_{\pi/6}^{\pi/2}\int_{2}^{4\sin\theta} r\,dr\,d\theta$$

$$=2\int_{\pi/6}^{\pi/2}\frac{r^2}{2}\Big|_{2}^{4\sin\theta}d\theta=\int_{\pi/6}^{\pi/2}(16\sin^2\theta-4)\,d\theta$$

$$16\cdot\frac{1-\cos2\theta}{2}$$

$$=8\theta-\frac{1}{2}\cdot\frac{\sin2\theta}{2}-4\theta=4\theta-8\frac{\sin2\theta}{2}\Big|_{\pi/6}^{\pi/2}=2\pi-\frac{4\pi}{6}+4\cdot\frac{\sqrt{3}}{2}$$

$$=\frac{4\pi}{3}+2\sqrt{3}$$

7. $x^2+y^2=4$ çemberinin dışında ve $x^2+y^2-4y=0$ çemberinin içinde kalan bölgenin alanı aşağıdakilerden hangisidir?

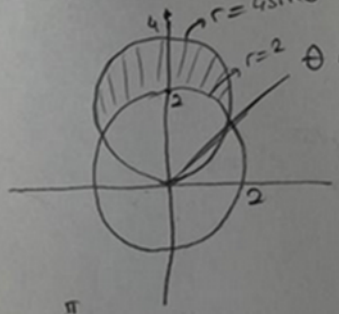
a) $\frac{1}{3}\pi+2\sqrt{3}$

b) $\frac{1}{2}\pi-2\sqrt{2}$

c) $\frac{4}{3}\pi+2\sqrt{3}$

d) $\frac{5}{3}\pi+3\sqrt{2}$

e) $\frac{2}{3}\pi-3\sqrt{5}$



$$A=2\int_{\pi/6}^{\pi/2}\int_{2}^{4\sin\theta} r\,dr\,d\theta=\int_{\pi/6}^{\pi/2}\frac{r^2}{2}\Big|_{2}^{4\sin\theta}d\theta=\int_{\pi/6}^{\pi/2}(8(1-\cos2\theta)-4)\,d\theta$$

$$=4\theta-4\sin2\theta\Big|_{\pi/6}^{\pi/2}=2\pi-\frac{2\pi}{3}+4\frac{\sqrt{3}}{2}$$

$$=\frac{4\pi}{3}+2\sqrt{3}.$$ bir2

9.

$$f(x,y) = \frac{1}{\sqrt{x^2+y^2}}$$

fonksiyonunun $a^2 \le x^2+y^2 \le b^2$, $0 < a < b$ bölgesi üzerindeki ortalama değeri aşağıdakilerden hangisidir?

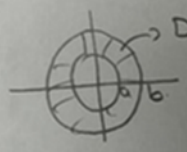
a) $\frac{2}{b-a}$

b) $\frac{3}{2(a+b)}$

c) $\frac{a+b}{b-a}$

d) $\frac{2}{b^2-a^2}$

e) $\frac{2}{a+b}$



→ D'nin alanı $= \pi(b^2-a^2)$

ort $f = \frac{1}{\text{D nin alanı}} \iint_D f(x,y)\,dA$

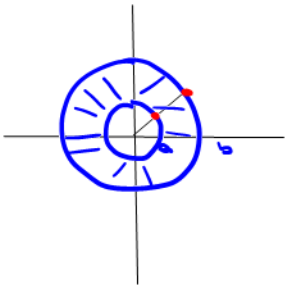
$= \frac{1}{\pi(b^2-a^2)} \int_0^{2\pi}\int_a^b \frac{1}{r} \cdot r\,dr\,d\theta$

$= \frac{1}{\pi(b-a)(b+a)} \cdot 2\pi \cdot (b-a) = \frac{2}{b+a}$

$f(x,y)$ nin O bölgesi üzerindeki ortalama değeri:

$$\bar{f} = \frac{\iint_O f(x,y)\,dx\,dy}{O\text{'nin Alanı}}$$

Daire Alanı $= \pi r^2$



→ O'nin Alanı $= \pi b^2 - \pi a^2 = \pi(b^2-a^2)$

$$\iint_O \frac{1}{\sqrt{x^2+y^2}}\,dx\,dy = \int\int_O \frac{1}{r} \cdot r\,dr\,d\theta = \int_0^{2\pi}\int_a^b dr\,d\theta$$

$$= \left(\int_0^{2\pi} d\theta\right)\left(\int_a^b dr\right)$$

$$= \theta\Big|_0^{2\pi} \cdot r\Big|_a^b = 2\pi(b-a)$$

$x = r\cos\theta$

$y = r\sin\theta$

$dx\,dy = r\,dr\,d\theta$

$x^2+y^2 = r^2$

$x^2+y^2 = b^2 \to r = b$

$x^2+y^2 = a^2 \to r = a$

$$\bar{f} = \frac{2\pi(b-a)}{\pi(b-a)(b+a)} = \frac{2}{b+a}$$

6.

$$\int_1^2 \int_0^{\sqrt{4-y^2}} \frac{xy}{\sqrt{x^2+y^2}}\, dx dy$$

integralinin kutupsal formu aşağıdakilerden hangisidir?

a) $\int_{\pi/3}^{\pi/2} \int_{\csc\theta}^{2} \frac{1}{2} r^2 \cos 2\theta \, dr d\theta$

b) $\int_{\pi/3}^{2\pi/3} \int_{\sec\theta}^{2} \frac{1}{4} r^2 \sin 2\theta \, dr d\theta$

c) $\int_{\pi/6}^{\pi/3} \int_{\sin\theta}^{2} \frac{1}{3} r^2 \sin 2\theta \, dr d\theta$

(d) $\int_{\pi/6}^{\pi/2} \int_{\csc\theta}^{2} \frac{1}{2} r^2 \sin 2\theta \, dr d\theta$

e) $\int_{\pi/6}^{\pi/2} \int_{2}^{\sec\theta} \frac{1}{2} r^2 \sin 2\theta \, dr d\theta$

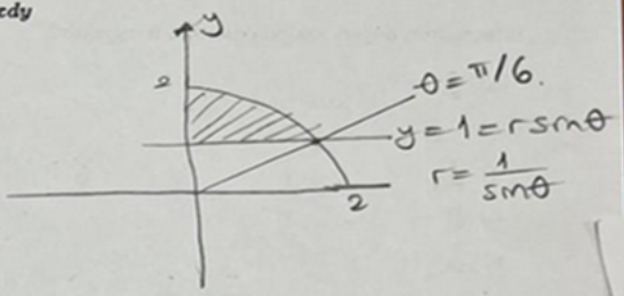
$y=1$, $y=2$

$x=0$, $x=\sqrt{4-y^2}$

$x = r\cos\theta$

$y = r\sin\theta$

$dxdy = r\, dr\, d\theta$



$r = 2 = \frac{1}{\sin\theta} \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_{\operatorname{cosec}\theta}^{2} \frac{r^2 \sin\theta \cdot \cos\theta}{r} \cdot r\, dr d\theta = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_{\operatorname{cosec}\theta}^{2} \frac{1}{2} r^2 \sin 2\theta \, dr d\theta$$

$$I = \int_1^2 \int_0^{\sqrt{4-y^2}} \frac{xy}{\sqrt{x^2+y^2}} dx dy \rightarrow \text{Kutupsal?}$$

$x = r\cos\theta$ $\quad x^2+y^2 = r^2$

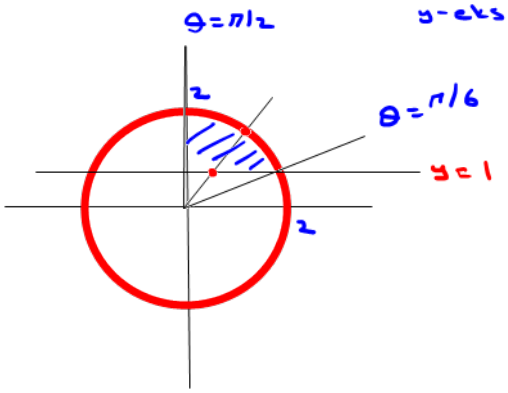
$y = r\sin\theta$

$dxdy = r\, dr d\theta$

D: $y=1$ $\quad y=2$ $\quad x=0$ $\quad x=\sqrt{4-y^2}$

$x=0$ → y-eks.

$x=\sqrt{4-y^2} \hookrightarrow x^2+y^2=4$



$y=1$ → $r\sin\theta = 1$ → $r = \frac{1}{\sin\theta}$

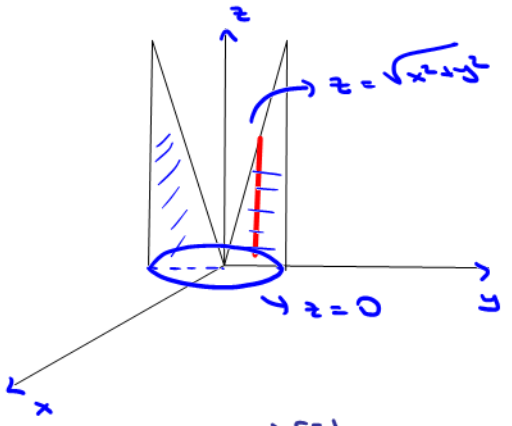
$x^2+y^2=4$ → $r^2=4$ → $r=2$

$2 = \frac{1}{\sin\theta} \rightarrow \sin\theta = \frac{1}{2} \rightarrow \theta = \frac{\pi}{6}$

$$I = \int_{\pi/6}^{\pi/2} \int_{1/\sin\theta}^{2} \frac{r^2 \cos\theta \sin\theta}{r} \cdot r\, dr d\theta = \int_{\pi/6}^{\pi/2} \int_{\operatorname{cosec}\theta}^{2} r^2 \underbrace{\cos\theta \sin\theta}_{\frac{\sin 2\theta}{2}} dr d\theta$$

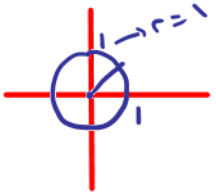
$$= \int_{\pi/6}^{\pi/2} \int_{\operatorname{cosec}\theta}^{2} \frac{r^2}{2} \sin 2\theta \, dr d\theta$$

(24) üstten koni $z=\sqrt{x^2+y^2}$ yüzeyi, altdan $z=0$ düzlemi, yandan silindir $x^2+y^2=1$ yüzeyi ile sınırlı cismin hacmini veren integral?



$$V=\iint_D (\text{yük.})\,(\text{T. Alanı}) \qquad \text{yük.} \downarrow \sqrt{x^2+y^2}, \quad \text{T. Alanı} \to dx\,dy$$

D: $x^2+y^2=1$ → r=1



$x=r\cos\theta$

$y=r\sin\theta$

$dx\,dy=r\,dr\,d\theta$

$x^2+y^2=r^2$

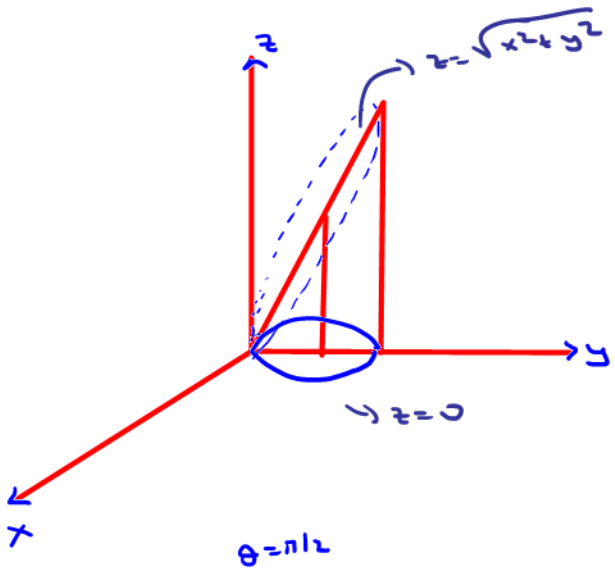
$$= \int_0^{2\pi}\int_0^1 r\cdot r\,dr\,d\theta = \int_0^{2\pi}\int_0^1 r^2\,dr\,d\theta$$

$$= \left[\int_0^{2\pi} d\theta\right]\cdot\left[\int_0^1 r^2\,dr\right] \qquad \int_0^1 r^2\,dr = \frac{r^3}{3}\Big|_0^1$$

$$= 2\pi\cdot\frac{1}{3} = \frac{2\pi}{3}$$

ötelenmiş silindir — koni

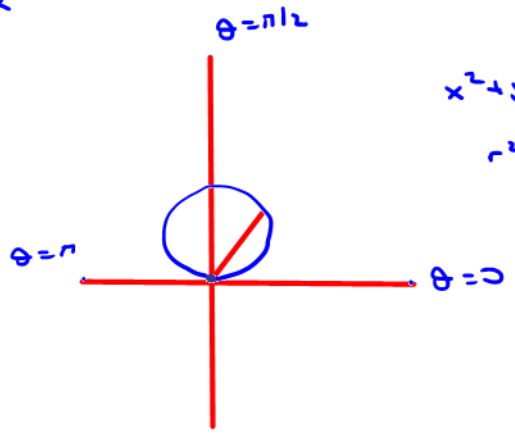
$x^2+y^2-4y=0$, $z=0$, $z=\sqrt{x^2+y^2}$ yüzeyler arasındaki cismin hacmini veren integral?



$$V=\iint_{D\ (yük.)}\ (T.Alanı)\ dx\,dy$$

$\sqrt{x^2+y^2}$ ↓

r , $r\,dr\,d\theta$



$x^2+y^2=4y$

$r^2=4r\sin\theta$

$r=4\sin\theta$

$x=r\cos\theta$

$y=r\sin\theta$

$dx\,dy=r\,dr\,d\theta$

$$V=\int_0^{\pi}\int_0^{4\sin\theta} r^2\,dr\,d\theta$$

$$=\int_0^{\pi}\frac{r^3}{3}\Big|_0^{4\sin\theta}\,d\theta$$

$$=\int_0^{\pi}\frac{64}{3}\sin^3\theta\,d\theta$$

$\sin^3\theta=\sin^2\theta\cdot\sin\theta$

$1-\cos^2\theta$

$$=\frac{64}{3}\int_0^{\pi}(1-\cos^2\theta)\sin\theta\,d\theta$$

$1-u^2$, $-du$

$$=-\frac{64}{3}\left[\cos\theta-\frac{\cos^3\theta}{3}\Big|_0^{\pi}\right]$$

$$=-\frac{64}{3}\left[-1-1+\frac{1}{3}+\frac{1}{3}\right]=-\frac{64}{3}\left[-2+\frac{2}{3}\right]=\frac{256}{9}$$

8. R, $y=-x$, $y=-x+1$, $y=3x$ ve $y=3x-1$ eğrileri ile sınırlı bölge olmak üzere

$$\iint_R \underbrace{(3x-y)}_{u}^{\frac{3}{2}} \underbrace{(x+y)}_{v}^{5}\, dA$$

integralinin değeri aşağıdakilerden hangisidir?

a) $\frac{1}{60}$
b) $-\frac{2}{25}$
c) $\frac{1}{20}$
d) $\frac{14}{41}$
e) $\frac{3}{28}$

$u=3x-y$
$v=x+y$

R	R'
$y+x=0$	$v=0$
$y+x=1$	$v=1$
$3x-y=0$	$u=0$
$3x-y=1$	$u=1$

$$J=\frac{1}{\begin{vmatrix}3 & -1\\ 1 & 1\end{vmatrix}}=\frac{1}{4}$$

$$=\frac{1}{4}\int_0^1\int_0^1 u^{3/2}v^5\,du\,dv=\frac{1}{4}\cdot\frac{2}{5}u^{5/2}\frac{v^6}{6}\Big|_0^1$$

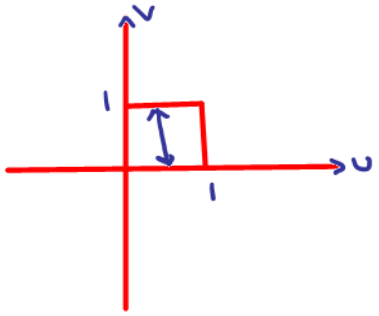
$$=\frac{1}{10}\cdot\frac{1}{6}=\frac{1}{60}$$

R: $y=-x$, $y=-x+1$, $y=3x$, $y=3x-1$

$$\iint_R \underbrace{(3x-y)}_{u}^{3/2}\cdot\underbrace{(x+y)}_{v}^{5}\,dA=?$$

① $u=3x-y$ $\quad v=x+y$

R		R'
$y+x=0$	→	$v=0$
$y+x=1$	→	$v=1$
$3x-y=0$	→	$u=0$
$3x-y=1$	→	$u=1$



② $dx\,dy=|J|\,du\,dv$

$$J=\frac{\partial(x,y)}{\partial(u,v)}=\frac{1}{\frac{\partial(u,v)}{\partial(x,y)}}$$

$$=\frac{1}{\begin{vmatrix}u_x & u_y\\ v_x & v_y\end{vmatrix}}=\frac{1}{\begin{vmatrix}3 & -1\\ 1 & 1\end{vmatrix}}=\frac{1}{4}$$

③ $$I=\iint_{R'} u^{3/2}\cdot v^5\cdot\frac{1}{4}\,du\,dv=\frac{1}{4}\int_0^1\int_0^1 u^{3/2}\cdot v^5\,dv\,du$$

$$=\frac{1}{4}\int_0^1 u^{3/2}\cdot\frac{v^6}{6}\Big|_0^1\,du$$

$$=\frac{1}{24}\int_0^1 u^{3/2}\,du=\frac{1}{24}\cdot\frac{u^{5/2}}{\frac{5}{2}}\Big|_0^1=\frac{1}{60}$$

$A = \iint_D dx\,dy$

(17) $D:\begin{cases} y=x^2 \\ y=2x^2 \\ x=y^2 \\ x=3y^2 \end{cases}$ bölgesinin alanını $u=\frac{y}{x^2}$ $v=\frac{x}{y^2}$ dönüşümü ile yazarsak integral?

a) $\int_1^2\int_1^3 \frac{1}{3u^2v^2}\,dv\,du$ ~~b)~~ $\int_1^2\int_1^3 dv\,du$ ~~c)~~ $\int_1^3\int_1^3 \frac{1}{3u^2v^2}\,du\,dv$

veren integral:

①

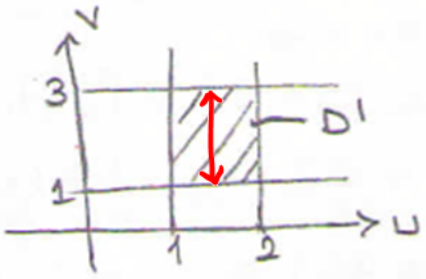
$u=\frac{y}{x^2}$

$v=\frac{x}{y^2}$

$u\cdot v=\frac{1}{xy}$

$xy=\frac{1}{uv}$

D	D'
$\frac{y}{x^2}=1$	$\rightarrow u=1$
$\frac{y}{x^2}=2$	$\rightarrow u=2$
$\frac{x}{y^2}=1$	$\rightarrow v=1$
$\frac{x}{y^2}=3$	$\rightarrow v=3$



③ $\frac{D(x,y)}{D(u,v)} = \frac{1}{\frac{D(u,v)}{D(x,y)}} = \frac{1}{\begin{vmatrix} -2y/x^3 & 1/x^2 \\ 1/y^2 & -2x/y^3 \end{vmatrix}} = \frac{x^2y^2}{3} = \frac{1}{3u^2v^2}$

$A = \iint_{D'} \frac{1}{3u^2v^2}\,du\,dv = \frac{1}{3}\int_1^2\int_1^3 \frac{1}{u^2v^2}\,dv\,du$

$J.du\,dv$

$A=\iint_D dx\,dy$

$f(x,y,z) = -\sqrt{x^2+z^2}$ ve C eğrisi parametrik olarak

$r(t) = (a\cos t)\,j + (a\sin t)\,k$, $0 \le t \le 2\pi$ veriliyor.

$(a>0)$

Buna göre $\int_C f(x,y,z)\,ds = ?$

$$\int_C f\,ds = \int_a^b f(r(t)).\,|\vec{r}'|.\,dt$$

$\vec{r} = \langle 0, a\cos t, a\sin t \rangle$ → $\vec{r}' = \langle 0, -a\sin t, a\cos t \rangle$

$x \downarrow \quad y \downarrow \quad z \downarrow$

$|\vec{r}'| = \sqrt{a^2\sin^2 t + a^2\cos^2 t} = \sqrt{a^2} = |a| = a$

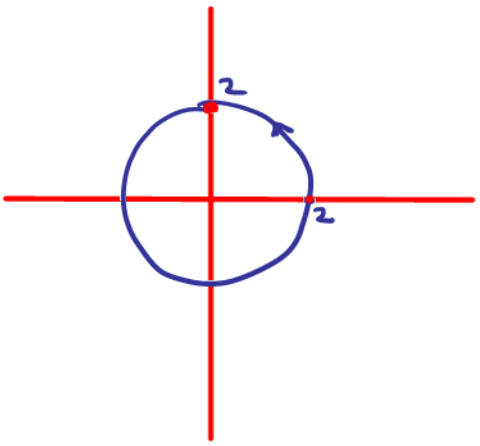
$f = -\sqrt{x^2+z^2} = -\sqrt{0^2+a^2\sin^2 t} = -a|\sin t|$

$$I = \int_0^{2\pi} -a|\sin t|.\,a\,dt = -a^2\left[\int_0^{\pi} \sin t\,dt + \int_{\pi}^{2\pi} -\sin t\,dt\right]$$

$$= -a^2\left[-\cos t\Big|_0^{\pi} + \cos t\Big|_{\pi}^{2\pi}\right]$$

$$= -a^2\left[1+1+1+1\right] = -4a^2$$

⑨ C eğrisi: $x^2+y^2=4$ çemberinin 1. Bölgesinde (2,0) noktasını (0,2) noktasına bağlayan bir eğri olsun. $f(x,y)=x+y$ nin C üzerindeki eğrisel integrali?



(2,0) $t=0$ (0,2) $t=\frac{\pi}{2}$

$x^2+y^2=4 \rightarrow x=2\cos t$, $y=2\sin t$

$$\int_C f\,ds = \int_0^{\pi/2} f(\vec{r}).|\vec{r}'|.dt$$

$\vec{r}=\langle 2\cos t, 2\sin t\rangle \rightarrow \vec{r}'=\langle -2\sin t, 2\cos t\rangle$

$|\vec{r}'|=\sqrt{4\sin^2 t+4\cos^2 t}=2$

$f=x+y=2\cos t+2\sin t=2(\cos t+\sin t)$

$$I=\int_0^{\pi/2} 4.(\cos t+\sin t)\,dt = 4\left[\sin t-\cos t\right]_0^{\pi/2} = 4[1+1]=8$$

Soru

C eğrisi; $z=\sqrt{x^2+y^2}$ konisi ile $z=1$ düzleminin arakesitindeki $(1,0,1)$ noktasından $(0,1,1)$ noktasına doğru saat yönünün tersi yönünde bir eğri olsun. $\vec{F}(x,y,z) = xz\,\vec{i} + (z^2+4)\,\vec{j} + (x^2+2yz)\,\vec{k}$ olsun.

$$\int_C \vec{F}\cdot d\vec{r} = ?$$

$$\int_C \vec{F}\cdot d\vec{r} = \int_a^b \vec{F}(\vec{r})\cdot\vec{r}'\,dt$$

(skaler çarpım)

$z=\sqrt{x^2+y^2} \iff z=1$

$C: x^2+y^2=1,\ z=1$

$x=\cos t$

$y=\sin t$

$(1,0,1)$ $t=0$ — $(0,1,1)$ $t=\frac{\pi}{2}$

$0\le t\le \pi/2$

$\vec{r} = \langle \cos t, \sin t, 1\rangle$

$\vec{r}' = \langle -\sin t, \cos t, 0\rangle$ ✱

$\vec{F} = \langle xz, z^2+4, x^2+2yz\rangle = \langle \cos t, 5, \cos^2 t + 2\sin t\rangle$ ✱

$\vec{F}\cdot\vec{r}' = -\sin t\cos t + 5\cos t$

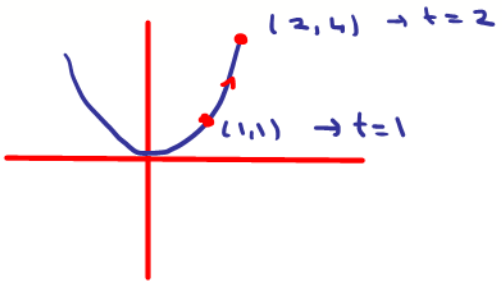
$$I = \int_0^{\pi/2} (-\underbrace{\sin t\cos t}_{\frac{\sin 2t}{2}} + 5\cos t)\,dt = \frac{\cos 2t}{4} + 5\sin t \Big|_0^{\pi/2}$$

$$= -\frac{1}{4} + 5 - \frac{1}{4} = 5 - \frac{1}{2} = \frac{9}{2}$$

(17) C eğrisi : $y = x^2$ üzerinde (1,1) noktasını (2,4) noktasına götüren eğri olmak üzere $\vec{F} = \langle x^2 - y, xy^2 \rangle$ vektör alanının C üzerindeki eğrisel integrali?

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}) \cdot \vec{r}' \cdot dt$$

sk. çarpım



$y = x^2 \rightarrow x = t \quad y = t^2$

C: $\vec{r} = \langle t, t^2 \rangle \qquad 1 \le t \le 2$

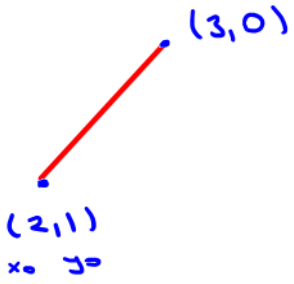
$\vec{r}' = \langle 1, 2t \rangle$

$\vec{F} \cdot \vec{r}' = 2t^4$ *

$\vec{F} = \langle x^2 - y, xy^2 \rangle = \langle t^2 - t^2, t^5 \rangle = \langle 0, t^5 \rangle$

$$I = \int_1^2 2t^4 \, dt = 2 \frac{t^7}{7}\Big|_1^2 = \frac{2}{7}\{128 - 1\} = \frac{254}{7}$$

(13) C: (2,1) noktasını (3,0) noktasına götüren doğru parçası ise $\int_C (x+y)^2\,dx + 2x\,dy = ?$



$\vec{v} = \langle 1, -1 \rangle$ (a, b)

$x = 2 + t$

$y = 1 - t$

$\vec{r} = \langle 2 + t, 1 - t \rangle$

$0 \le t \le 1$

$(2,1) \to t = 0$

$(3,0) \to t = 1$

$x = 2 + t \to dx = dt$ $\quad y = 1 - t \to dy = -dt$

$$I = \int_0^1 (2 + t + 1 - t)^2 \cdot dt + 2 \cdot (2 + t) \cdot (-dt)$$

$$= \int_0^1 [9 - 4 - 2t]\,dt = \int_0^1 (5 - 2t)\,dt = 5t - t^2 \Big|_0^1 = 5 - 1 = 4$$

C: $(0,0,0) \xrightarrow{C_1} (1,0,1) \xrightarrow{C_2} (0,1,2)$

C_1: $x = (1-0)\cdot t + 0 = t \to dx = dt$
$y = (0-0)\cdot t + 0 = 0 \to dy = 0$
$z = (1-0)t + 0 = t \to dz = dt$

C_2: $x = (0-1)\cdot t + 1 = 1 - t \to dx = -dt$
$y = (1-0)\cdot t + 0 = t \to dy = dt$
$z = (2-1)t + 1 = t + 1 \to dz = dt$

3. C eğrisi $(0,0,0)$ noktasından $(1,0,1)$ noktasına ve $(1,0,1)$ noktasından $(0,1,2)$ noktasına olan doğru parçalarının birleşimi olmak üzere

$$\int_C (y+z)dx + (x+z)dy + (x+y)dz = \int_{C_1} + \int_{C_2}$$

eğrisel integralinin değeri aşağıdaki şıklardan hangisinde verilmiştir?

a) 1
b) $\frac{2}{3}$
c) $\frac{1}{4}$
d) $\frac{1}{6}$
e) 2

$\int_0^1 (t\,dt + 2t\cdot 0 + t\,dt) + \int_0^1 (2t+1)(-dt) + 2\,dt + 1\,dt$
$-2t - 1 + 2 + 1$
$t^2\Big|_0^1 + (-t^2) + 2t\Big|_0^1$
$= 1 - 1 + 2 = 2$

C_1: $(0,0,0) \to (1,0,1)$ doğru:

$\vec{v} = \langle 1,0,1 \rangle$

$0 \le t \le 1$

$x = t \to dx = dt$
$y = 0 \to dy = 0$
$z = t \to dz = dt$

$$I_1 = \int_0^1 t\,dt + t\,dt = \int_0^1 2t\,dt = t^2\Big|_0^1 = 1$$

C_2: $(1,0,1) \to (0,1,2)$
$x_0\ y_0\ z_0$

$\vec{v} = \langle -1,1,1 \rangle$
$a\ b\ c$

$0 \le t \le 1$

$x = 1 - t \to dx = -dt$
$y = t \to dy = dt$
$z = 1 + t \to dz = dt$

$$I_2 = \int_0^1 \underbrace{(1+2t)(-dt)}_{-dt-2t\,dt} + \underbrace{2\,dt + 1\,dt}_{3\,dt} = \int_0^1 (2 - 2t)\,dt = 2t - t^2\Big|_0^1 = 1$$

$$I = I_1 + I_2 = 1 + 1 = 2$$

(*) $\lim_{n\to\infty} a_n = 10$ ve $\lim_{n\to\infty} b_n = 2$ olsun.

$\lim_{n\to\infty} \frac{a_n + n^2 b_n}{2n^2+1} = ?$ a) 2 b) $\frac{1}{2}$ c) 10 d) 11 e) 12

$\lim_{n\to\infty} \frac{\overbrace{a_n}^{10} + n^2 \cdot \overbrace{b_n}^{2}}{2n^2+1} \to \frac{2n^2+10}{2n^2+1} \xrightarrow{n\to\infty} \frac{2}{2} = 1$

(7) $a_1 = \frac{1}{2}$ ve $n \geq 1$ doğal sayısı için $a_{n+1} = \sqrt{3+a_n} - 1$ ile verilen $\{a_n\}$ dizisinin limitinin 1 olduğu bilindiğine göre $\left\{ \frac{a_{n+1} - 1}{a_n - 1} \right\}$ dizisinin limiti?

$\lim_{n \to \infty} a_n = 1$

$$\lim_{n \to \infty} \frac{a_{n+1} - 1}{a_n - 1} = \lim_{n \to \infty} \frac{\sqrt{3+a_n} - 2}{a_n - 1} \to \frac{0}{0}$$

($\sqrt{3+a_n} \to 1$)

$$= \lim_{n \to \infty} \frac{3 + a_n - 4}{(a_n - 1) \cdot (\sqrt{3+a_n} + 2)} = \frac{1}{2+2} = \frac{1}{4}$$

($\sqrt{3+a_n} \to 1$)

(*) $\frac{2n^2+4n^3+3}{1+2n^3} < a_n < \frac{2n+1}{n+1}$ $(\forall n \in \mathbb{Z}^+)$

$\{b_n\} = \left\{\cos\left[(2n+1)\frac{\pi}{2}\right]\right\}$ olsun. $\{a_n\}$ ve $\{b_n\}$ için hangileri doğrudur!

$\frac{4n^3+2n^2+3}{1+2n^3} < a_n < \frac{2n+1}{n+1}$

$\downarrow$ $\lim = 2$; $\downarrow$ $\lim = 2$

↳ Sıkıştırma Teo. $\Rightarrow \lim_{n\to\infty} a_n = 2$

$\Rightarrow \{a_n\}$ yakınsak

I. $\{a_n\}$ yakınsaktır ✓

II. $\{b_n\}$ ıraksaktır ✗

III. $\lim_{n\to\infty} b_n$ mevcut değildir. ✗

IV. $\lim_{n\to\infty} a_n = 2$ ✓

V. $\lim_{n\to\infty} (a_n \cdot b_n) = 0$ ✓

$\{b_n\} = \left\{\cos\frac{(2n+1)\pi}{2}\right\}_{n=1}^{\infty} = \{0, 0, 0, \dots, 0, \dots\}$ → Dizi 0'a yığılır

$\lim_{n\to\infty} b_n = 0$

$\Rightarrow$ Dizi yakınsak

$\lim_{n\to\infty} (a_n \cdot b_n) = 0$

$\lim_{n\to\infty} a_n = 2$, $\lim_{n\to\infty} b_n = 0$